

Savitribai Phule Pune University				
Third Year of Mechanical Engineering (2024 Pattern)				
PCC303-MEC: Design of Machine Elements				
Teaching Scheme		Credit	Examination Scheme	
Theory	4 Hours/Week	4	CCE	30 Marks
Practical	NA		End-Semester	70 Marks
Prerequisite Courses, if any: The basics of material elastic behavior, stress, strain, its relationship, failure modes, different theories of failure and its applications. The design cycle, basis of design considerations like strength, rigidity, manufacture, assembly and cost, standards and codes. The preferred sizes and series, tolerances and types of fits. Construction of SMD and BMD. Roots of equations, Interpolation rule.				
Course Objectives: 1. UNDERSTAND various design considerations, design procedures. 2. SELECT materials for a specific application. 3. EVALUATE the stresses in machine components due to various types of loads and failure. 4. DESIGN various machine components such as shafts, couplings, keys, screws, joints, springs. 5. ANALYZE machine components subjected to variable loading for finite and infinite life.				
Course Outcomes: After successful completion of the course, learner will be able to: CO. 1 DESIGN levers and components subjected to eccentric loading by selecting appropriate material. CO. 2 DESIGN shafts, keys and couplings under different loading conditions. CO. 3 EVALUATE the stresses developed in the different type of welded and threaded joints. CO. 4 APPLY the design procedure for different types of springs and EVALUATE dimensions of machine components under fluctuating loads. CO. 5 ANALYZE different stresses developed in power screws and APPLY those in the procedure to design screw jack.				
Course Contents				
Unit I	Fundamentals of Machine Design and Material Selection			(07 Hours)
Selection of materials and factor of safety, Service factor, Design of hand / foot lever, lever for safety valve, bell crank lever, Design of components subjected to eccentric loading., Design process , Design considerations, Engineering materials and their properties, Stress concentration, Static and fluctuating loads.				
Real World Assignment 1 Determine whether an elevator cable is safe under the maximum passenger load. 2 Determine the factor of safety of motorcycle side stand and verify whether design is safe. 3 Identification of different engineering materials used in various parts of two wheeler with reasoning 4 Ergonomic Lever Design for Hand Lever or Foot Lever for bore-well pump 5 The Material selection and analysis of failed mechanical component such as snapped bolt, fan blade etc. 6 Stress Concentration for plate with a central hole				
Exemplar / Practical Applications Material Selection in Industry, Bell Crank Lever for Bicycle Braking Systems, Lever for Safety Valve for Industrial Boilers, Eccentric Loading in C-Clamps, Fluctuating Loads in Automobile Leaf Springs, Service Factor for different mechanical systems.				
Unit II	Design of Shafts, Keys and Couplings			(07 Hours)

Shaft design on the Strength basis, torsional rigidity basis and lateral rigidity basis, Design of shaft as per A.S.M.E. code. Design of square and rectangular keys, Kennedy key and splines. Design of Flange Coupling and Bushed-Pin Flexible Coupling.

Real World Assignment

- 1 Design of shaft for real world applications like lathe, pump, automobile etc.
- 2 Design, and analyze keys used in real mechanical systems for torque transmission.
- 3 Identify different types of couplings used in the transmission system and justify their use.

Exemplar / Practical Applications

- 1) Automotive engines and transmissions
- 2) Industrial machinery like conveyor systems and pumps
- 3) Agricultural equipment, including tractors and harvesters
- 4) Automotive powertrains in electric vehicles

Unit III

Design of Mechanical Joints

(08 Hours)

Introduction to threaded joints, Bolts of uniform strength, locking devices, eccentrically loaded bolted joint in shear, Eccentric load perpendicular and parallel to axis of bolt, Eccentric load on circular base. Applications of joints in mechanical systems

Introduction to welded joints, Strength of butt, parallel and transverse fillet welds, axially loaded unsymmetrical welded joints, Eccentric load in plane of welds, Welded joints subjected to bending and torsional moments. Design of cotter and knuckle joints

Real World Assignment

1. Industrial Visit to workshop
- 2 Design a cotter joint for bicycle so as to transmit power from pedal to axle of bicycle.
3. Design a knuckle joint for tractor and trailer so that tractor can pull the trailer of 3 ton. (Empty weight 0.5 ton + material weight 2.5 ton).

Exemplar / Practical Applications

Steam Engine -connecting piston rod to cross head, Pump-piston rod to tail pump rod (Cotter joint), automotive steering mechanism, train coupling, tractor -trolley linkage (knuckle joint), Pressure Vessels, pipe lines, automotive frames (welded joints).

Unit IV

Design of Springs and Mechanical Components under Variable Loads

(08 Hours)

Springs: Types and applications of springs, Stress and deflection equations for helical compression Springs, Springs in series and parallel, Design of helical springs, concentric helical springs, surge in spring, Design of Multi-leaf springs, Nipping of Leaf springs, Shot Peening.

Analysis of machine elements under variable loads: Fatigue Failure, Stress-Life (S-N) Curve and Endurance Limit, Factors Affecting Endurance Strength (Modifying Factors and Notch Sensitivity), Reversed Stress Concept, Fatigue Failure Theories: Soderberg, Goodman, and Gerber Criteria, Modified Goodman Diagram for Design Applications, Design for Finite and Infinite Life, Cumulative Damage in Fatigue, Fatigue Design Under Combined Stresses (Theoretical Approach)

Real World Assignment

- 1 Design a helical compression spring for a suspension system and analysis of Compression Spring Used in Motorcycle Suspension
- 2 study construction and verify basic design calculations of leaf springs.
- 3 Comparative Study of Helical Spring for various applications
- 4 Suspension system in automobile (Two wheeler / Car)
- 5 Fatigue Design of an Automotive Drive Shaft
- 6 Fatigue Analysis of a Connecting Rod

Exemplar / Practical Applications

Design of Helical Compression Spring for Automotive Suspension, Design of Multi-Leaf Spring for Light Commercial Vehicle, Fatigue Design of Rotating Shaft Under Variable Loading, Fatigue Life Estimation Using Cumulative Damage (Miner's Rule)

Unit V**Design of Power Screw****(08 Hours)**

Terminology of Power Screw, Torque analysis and Design of power screws with square and trapezoidal threads, Collar friction torque, Self-locking screw, Efficiency of square threaded screw, Efficiency of self-locking screw, Design of screw, nuts and C-Clamp. Design of screw jack, Differential and Compound Screw and Re-circulating Ball Screw **(Theoretical treatment only).**

Real World Assignment

- 1 Torque Requirement and Efficiency Estimation of a Bench Vice Screw Mechanism
- 2 Analysis of C-Clamp Screw and Collar Friction Effects
- 3 Design a Power screw for a Hydraulic Press
- 4 Design of Automotive steering gear (Recirculating Ball Type)
- 5 Design of a 2-Ton Mobile Scissor Lift Power Screw

Exemplar / Practical Applications

The car Jack, machine vise, Large mechanical Press.

Learning Resources**Text Books:**

- 1 Bhandari V.B., Design of Machine Elements, Tata McGraw Hill Publication Co. Ltd.
- 2 Shigley J.E. and Mischke C.R., Mechanical Engineering Design, McGraw Hill Publication Co. Ltd.

Reference Books:

- 1 Spotts M.F. and Shoup T.E., Design of Machine Elements, Prentice Hall International.
- 2 Juvinal R.C., Fundamentals of Machine Components Design, John Wiley and Sons.
3. Black P.H. and O. Eugene Adams, Machine Design, McGraw Hill Book Co. Inc.
4. William C. Orthwein, Machine Components Design, West Publishing Co. and Jaico Publications House.
5. Hall A.S., Holowenko A.R. and Laughlin H.G, Theory and Problems of Machine Design, Schaum's Outline Series.
6. C. S. Sharma and Kamlesh Purohit, Design of Machine Elements, PHI Learning Pvt. Ltd.
7. D. K. Aggarwal & P. C. Sharma, Machine Design, S.K Kataria and Sons.
8. P. C. Gope, Machine Design: Fundamentals and Applications, PHI Learning Pvt. Ltd.
9. Design Data - P.S.G. College of Technology, Coimbatore.
10. K. Mahadevan, K. Balveera Reddy, Design Data Handbook for Mechanical Engineers, CBS Publishers.

MOOC / NPTEL/YouTube Links: -

<https://www.youtube.com/watch?v=ofmbhbVCUqI&list=PL3D4EECEFAA99D9BE&index=3>

<https://www.youtube.com/watch?v=SL21aDqgs8Q>

<https://www.youtube.com/watch?v=TPURJnlekeo>